

Ex.No:1: Simulation of FDMA in MATLAB Date.

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Program:

```
clc; clear; close all;

fs = 1000;

t = 0:1/fs:1;

TBW = 600;

Users = 3;

CBW = TBW/Users;

fc = [100 300 500]; % Carrier frequencies

m = sin(2*pi*10*t); % Message signal

% FDMA signals for each user

s1 = m .* cos(2*pi*fc(1)*t);

s2 = m .* cos(2*pi*fc(2)*t);

s3 = m .* cos(2*pi*fc(3)*t);

% Plot FDMA channel allocation

subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'), grid on

title('FDMA Channel Allocation')

xlabel('Time (s)')

ylabel('Amplitude')

% Plot channel bandwidth per user

subplot(2,1,2)
```

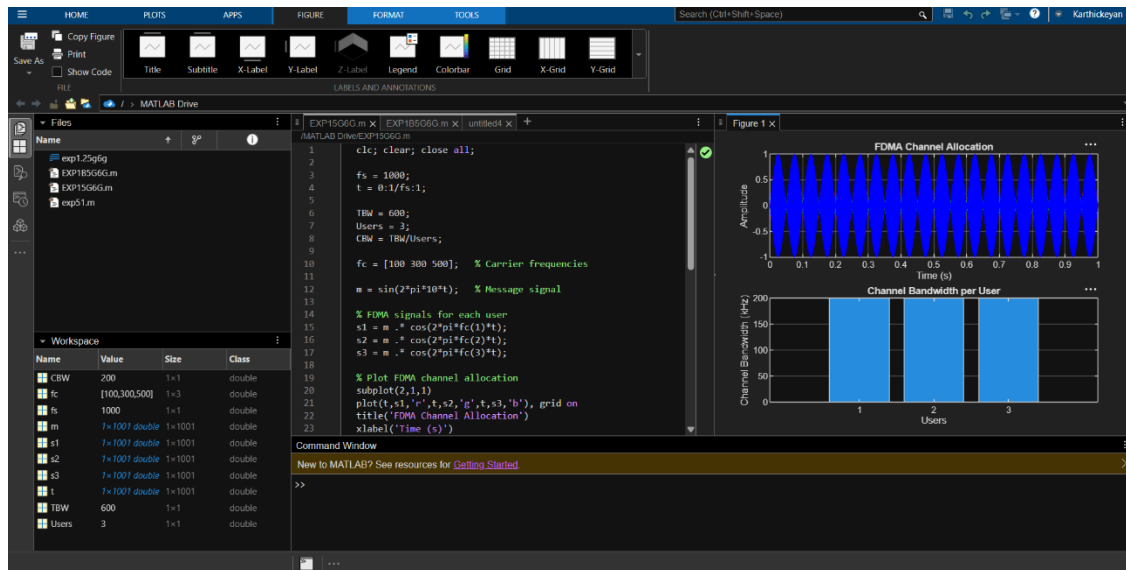
```
bar(1:Users, CBW*ones(1,Users)), grid on
```

```
xlabel('Users')
```

```
ylabel('Channel Bandwidth (kHz)')
```

```
title('Channel Bandwidth per User')
```

OUTPUT:



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated

channel bandwidth.

Program:

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

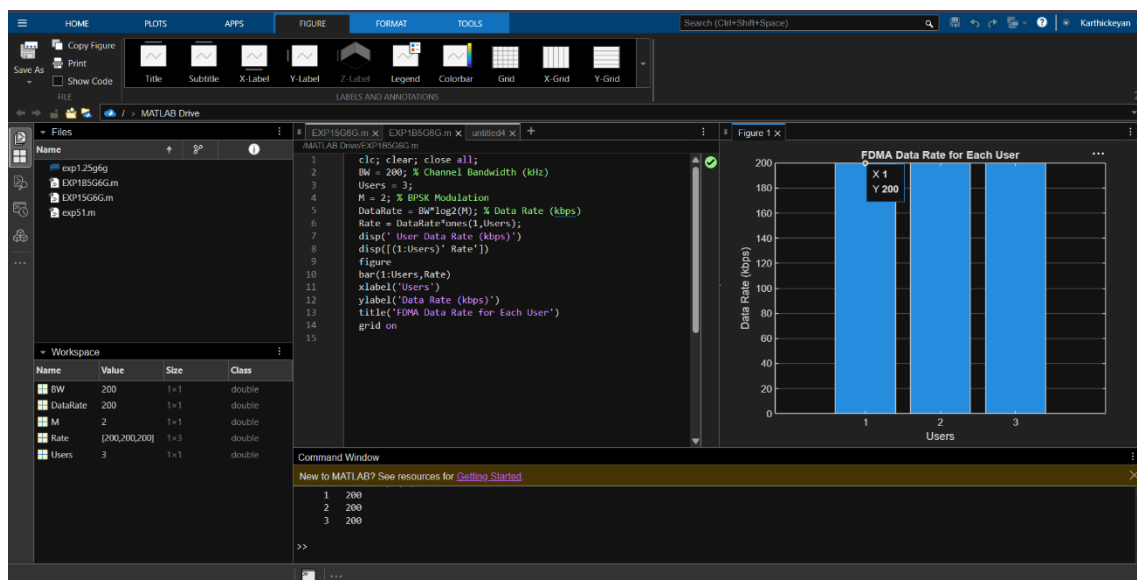
```
Rate = DataRate*ones(1,Users);
```

```

disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on

```

OUTPUT:



Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to

Noise Ratio (SNR) for different users.

Program:

```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```

SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on

```

OUTPUT:

